LitPro: Self-Improving Prompt Agents for Scientific Data Extraction

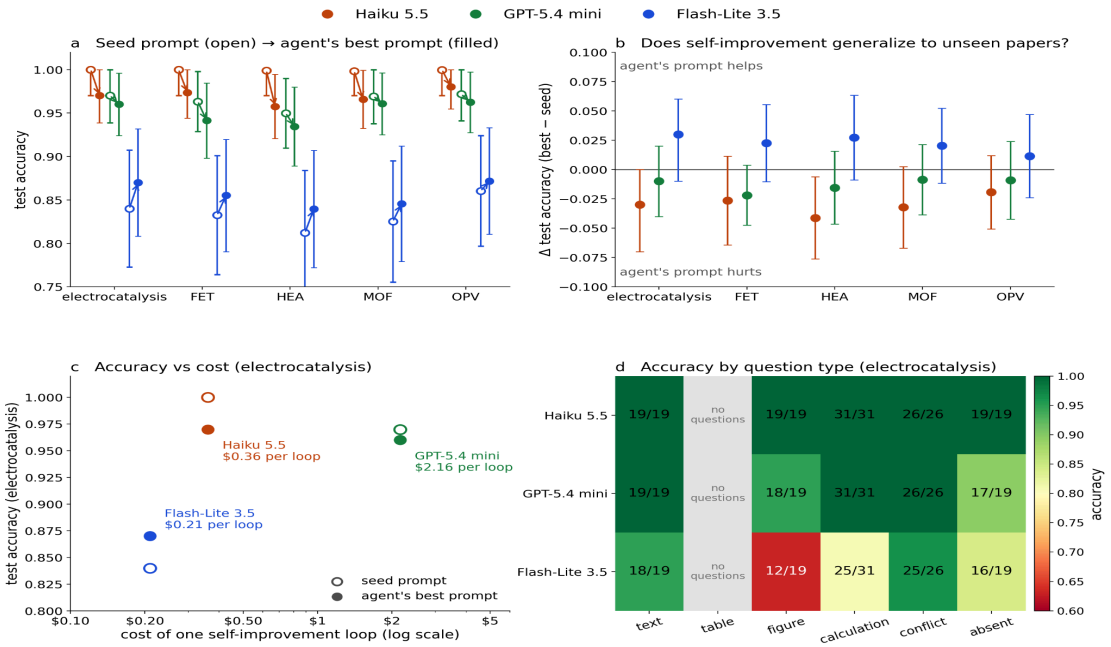
Rodrigo Ferreira (rpferreira@uchicago.edu), Ray Zhu (rzhu16@uchicago.edu)

Goal: test whether low-cost LLM agents can improve their own extraction prompts for chemistry and matsci papers, using only a score as feedback. For each domain, we measure accuracy v. cost/runtime & establish what a "good" prompt roughly looks like.

Data and questions: 269 open-access highly-cited, high-impact papers in 9 domains (~30/domain).1,614 questions (6 per paper), one of each type: a value from running text, a table, a figure, a calculation from reported values, a choice between conflicting values, and "absent" (no answer). All drafted by GPT-5.5 and verified by Claude Sonnet 5.5 and Gemini 3.6 Flash.

Method: agents are Claude Haiku 5.5, GPT-5.4 mini,Gemini 3.5 Flash-Lite. Each agent starts from a minimal prompt and rewrites it up to 5 times, seeing only its score on 10 training papers. It may change wording, structure, or partitioning (all questions in one call v. one call per question). The best prompt is then scored on 20 held-out papers with a $5/loop budget. Numeric answers are checked in code against a tolerance,text answers go to a panel of frontier judges (Claude Opus 5.5, GPT-5.5, Gemini 3.1 Pro), decided by majority vote.

Results:



1. **Tasks are near saturation:** with a 21-word seed prompt, even the cheapest models answer 84-100% of questions on unseen papers. The remaining errors are mostly figure-only values, calculations and "not reported" questions.
2. **Self-improvement doesn't yet generalize:** the agents explore sensibly (rewording, domain checklists, per-question calls) and gain 1-3 questions on the training papers. On unseen papers the change is within ±3 points, with every confidence interval including zero: with score-only feedback and 10 training papers, they overfit to noise.
3. **Cost separates the models, not accuracy.** Haiku gives the best accuracy at $0.36 per loop, GPT-5.4 mini costs 6x more with no gain, and Flash-Lite is the cheapest but least accurate. With caching, per-question calls cost only 1.4x more than one call per paper.

Next steps: run more domains, give agents per-question-type scores to curb overfitting, add harder tasks (multi-figure, cross-paper), and test prompt transfer across models and domains.